

UNIT 1 · TOPIC 1.9

Understanding Airline Schedules

Behind every departure time is a giant puzzle of planes, crews and slots.

B.B.A. Aviation Management · Semester III · Unit 1: Overview of the Airline Industry
Air Ticketing & Fare Construction (2ACBAM32) · Arun Kumar

**BOARDING PASS**

LEARNER

STU

PROFESSIONAL

**PRO**

FLIGHT

ATF 1.9

GATE

PERIOD 9

DURATION

00:40

SEAT

SEM III

CLASS

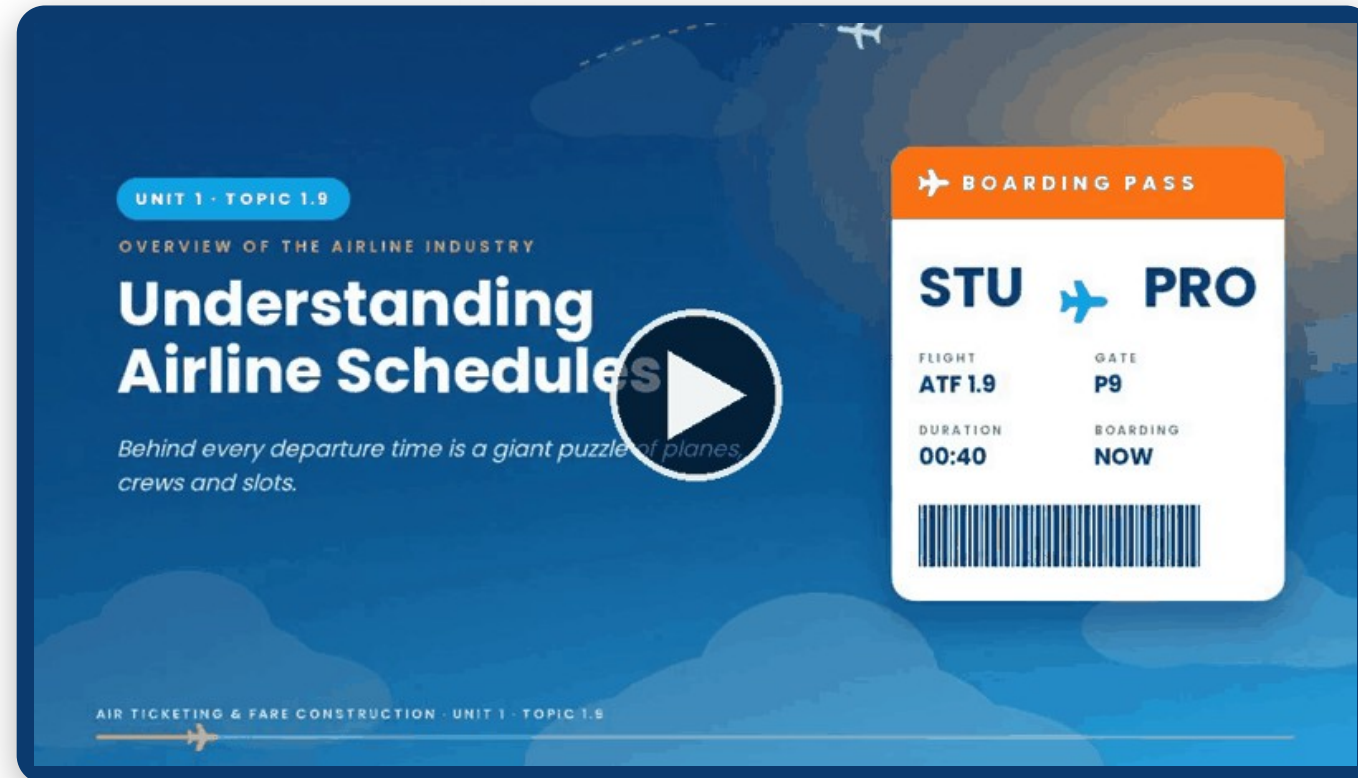
**PROBLEM-
SOLVING**

BOARDING

NOW

TOPIC 1.9

In-Flight Movie: Topic Intro



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02

TOPIC 1.9

Pre-Flight Question



THINK ABOUT IT

Your flight leaves at exactly 07:00. Who decided that time — and how long ago?

**THINK**

30 seconds alone

**PAIR**

1 minute with a neighbour

**SHARE**

2-3 answers with the class



TOPIC 1.9

Today's Flight Plan

1**1.9.1**

Scheduling fundamentals: how airlines determine flight schedules

2**1.9.2**

Scheduling challenges: weather, airport slots, maintenance and demand forecasting

3**1.9.3**

Types of schedules: block time, flight segments and aircraft routing

4**1.9.4**

How schedules impact ticket pricing and operations

**BY LANDING, YOU WILL**

- Describe the schedule-planning process
- Identify constraints that shape schedules
- Calculate block time and read segments
- Link departure times to fares and operations

Pedagogy: Learning Through Problem-Solving

TOPIC 1.9.1

How an Airline Builds Its Schedule

Planning starts up to a year before the first flight.



TOPIC 1.9.1

Scheduling Fundamentals

**Seasons**

Schedules are planned for IATA Summer and Winter seasons.

**Utilisation**

Aircraft earn money only when flying — LCCs aim for 12+ block hours a day.

**Turnaround**

Time on ground between flights: ~25-45 min for narrow-bodies.

**KEY IDEA**

A schedule is a trade-off: what passengers want (convenient times), what the aircraft can do (range, turnaround) and what the rules allow (slots, crew duty limits).



TOPIC 1.9.2

Scheduling Challenges

Problem: how do you keep 1,000+ daily flights on time?

**Weather**

North India winter fog, monsoon storms and cyclones disrupt operations.

**Airport Slots**

Busy airports like Delhi and Mumbai limit take-off/landing times.

**Maintenance**

Planned checks take aircraft out of service; unplanned faults cause delays.

**Demand Forecasting**

Festivals, holidays and events shift demand quickly.



TOPIC 1.9.2

Airport Slots Explained



Level 1

Non-coordinated: capacity meets demand; no slots needed.



Level 2

Schedule-facilitated: airlines adjust times voluntarily.



Level 3

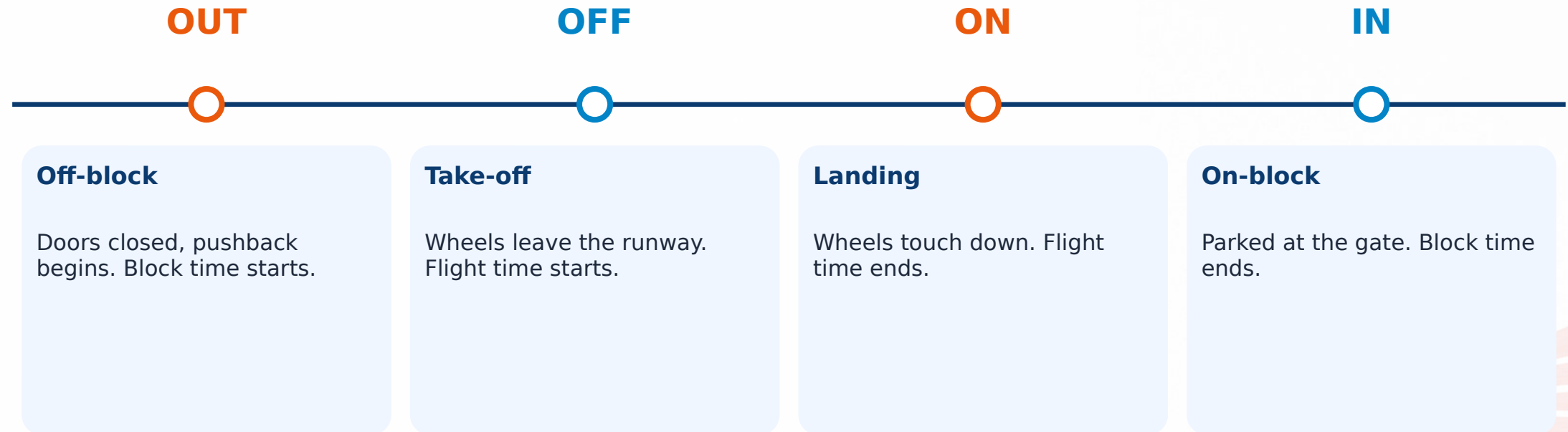
Fully coordinated: a slot is required to operate (e.g., Delhi, Mumbai, London Heathrow).



TOPIC 1.9.3

Block Time vs Flight Time

Block time runs from “chocks off” to “chocks on” — the time printed on schedules.



TOPIC 1.9.3

Segments, Legs & Aircraft Routing

One aircraft, one day, many flights.

- Leg: one take-off to one landing.
- Segment: what the passenger books from boarding to exit — may include a technical stop.
- Aircraft routing (tail assignment): the sequence of legs one aircraft flies in a day.
- A delay early in the rotation ripples through every later flight.



AIRCRAFT ROUTING · VT-XXX (illustrative)

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06:00 DEL-BOM 6E xx1
08:50 BOM-BLR 6E xx2
11:30 BLR-CCU 6E xx3
15:10 CCU-DEL 6E xx4
18:30 DEL-GOI 6E xx5
21:40 GOI-DEL 6E xx6
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// ~14 block hrs in one day
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TOPIC 1.9.4

How Schedules Shape Ticket Prices



Premium Timings

- Early-morning and evening business flights
- Friday evening / Sunday night
- Non-stop and short connections
- Usually higher fares

VS



Off-Peak Timings

- Mid-day and red-eye flights
- Tuesday / Wednesday departures
- Long connections or one-stop routes
- Usually lower fares



TOPIC 1.9.4

Schedules & Operations: The Ripple Effect



Crew Planning

Rosters follow the schedule and duty-time limits.



Maintenance Windows

Night stops at bases allow checks without cancelling flights.



Ground Handling

Staff and equipment planned around arrival waves.



Schedule Changes

Agents must reaccommodate passengers and reissue tickets when times change.



TOPIC 1.9

Cabin Crew Activity: Fog Day Crisis

**PEDAGOGY****Learning Through Problem-Solving****TIME · 12 min****FORMAT · Groups of 4****1**

Each group gets a one-day routing for two aircraft and a 90-minute fog delay at Delhi.

2

Identify all affected flights and passengers with tight connections.

3

Propose a recovery plan: swap aircraft, delay, cancel or merge flights.

4

Present your plan and explain its impact on passengers and cost.



TOPIC 1.9*Click to reveal each answer*

Pre-Landing Check

Q1**Block time starts when the aircraft...**

- A) Takes off
- B) Pushes back from the gate
- C) Lands
- D) Opens its doors

ANSWER**B****Q2****A Level 3 airport is...**

- A) Non-coordinated
- B) Schedule-facilitated
- C) Fully slot-coordinated
- D) Closed at night

ANSWER**C****Q3****The IATA Winter season starts on the last Sunday of...**

- A) September
- B) October
- C) November
- D) December

ANSWER**B****Q4****Which flight is usually cheapest?**

- A) Monday 07:00
- B) Friday 19:00
- C) Wednesday red-eye
- D) Sunday 21:00

ANSWER**C**

SAFE LANDING

Key Takeaways

- ✓ Schedules are planned by season from demand, fleet, crew and slots
- ✓ Weather, slots, maintenance and demand are the key challenges
- ✓ Block time = off-block to on-block; aircraft routing links legs
- ✓ Departure times directly influence fares and operations

EXIT TICKET Write 1 thing you learned today + 1 question you still have.

NEXT DEPARTURE**Topic 1.10 — Customer Service in Air Ticketing**

Thank you · Questions?